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Taylor series expansion of functions. 1

Taylor series expansion of functions.

```
% shushank

close all;
clc;
syms x; % define x as an symbolic variable

% define the function for which you want to compute the taylor series
f=sin(x);

% compute the taylor expansion on the point 'a' up to order 'n'
a=0; %The point around which you want to expand
n=10;%The order of the Taylor series expansion

% Compute the taylor series
taylor_seires=taylor(f,x,a,'order',n);

%Display taylor series
disp('taylor series Expansion');
disp(taylor_seires);
```

```
taylor series Expansion
x^9/362880 - x^7/5040 + x^5/120 - x^3/6 + x
```

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